

## Term & Condition

1. Agreement and consent: By submitting this form, I acknowledge and agree to comply with the stated rules and regulations of the hackathon, ensuring the accuracy and truthfulness of the provided information. I affirm that the innovation being submitted for Tata Technologies InnoVent Hackathon is solely conceptualized by our team, and we hold the necessary rights and permissions to submit and present this idea. I understand and commit to adhere to the InnoVent Privacy Policy notice, Tata Technologies privacy policy and the terms and conditions outlined for participation. I also consent to the capture and storage of the provided contact details for communication and administrative purposes related to the hackathon.

\*

InnoVent privacy notice: <https://www.tatatechnologies.com/in/innovent-privacy-notice/> Rules and regulations: <https://www.tatatechnologies.com/in/tata-technologies-innovent-rules-and-regulations/> Terms and conditions: <https://www.tatatechnologies.com/in/tata-technologies-innovent-terms-and-conditions/> Tata Technologies privacy policy: <https://www.tatatechnologies.com/in/privacy-policy/>



I have read and agree to the InnoVent privacy notice, Rules and Regulations, Terms and Conditions and Tata Technologies Privacy Policy

## Team Registration

### 2. Team Name \*

Add a creative and unique name that represents your team's identity and spirit.

NEC\_IT\_INNOVATORS

### 3. Team Leader Name \*

Enter the name of the designated team leader who will be the primary point of contact for the hackathon.

Sharumathi M

### 4. Team Leader Email Id \*

This Email Id will be used for all further communication and updates related to the hackathon

24205035@nec.edu.in

5. Team Leader Gender \*

Female



6. Team Leader Contact Number \*

Provide the 10 digit contact number of the team leader for communication purposes during the hackathon.

9360689731

7. Select College Country (Country where your college is located)? \*

India



8. College State (State where your college is located)? \*

Tamil Nadu



9. Team Leader Institute Name \*

Provide the name of the institute and the city

National Engineering College

10. College City (City where your college is located). \*

Kovilpatti-628503.

11. Team Leader Engineering Branch \*

Select the engineering branch in which the team leader specializes from the dropdown list provided. If the branch is not listed, please choose "Other" and specify the branch in the provided text field.

Information Technology



12. Team Leader Graduating Year \*

Select the anticipated graduating year of the team leader from the dropdown options

2028



13. Training & Placement Officer Name

Enter the name of the Training and Placement Officer

Enter your answer

14. Training & Placement Officer Email ID

Provide the email address for the Training and Placement Officer

Enter your answer

15. Faculty Advisor Name

Add the name of the faculty advisor who will be involved with your team

Enter your answer

16. Faculty Advisor Email ID

Provide the email address for the faculty advisor

Enter your answer

17. Do you have team members? \*

Yes



## 1st Team Member Registration

18. Team Member 1 Name \*

19. Team Member 1 Email Id

\*

24205031@nec.edu.in

20. Team Member 1 Gender \*

Female



21. Team Member 1 Contact number

\*

9944065128

22. Team Member 1 Engineering Branch \*

Information Technology



23. Team Member 1 Graduating Year \*

2028



## 2nd Team Member Registration

24. Team Member 2 Name

Bala Nithya M

25. Team Member 2 Email Id

26. Team Member 2 Gender

Female



27. Team Member 2 Contact Number

9344726227

28. Team Member 2 Engineering Branch

Information Technology



29. Team Member 2 Graduating Year

2028



### 3rd Team Member Registration

30. Team Member 3 Name

Siva Sankari R

31. Team Member 3 Email Id

24205028@nec.edu.in

32. Team Member 3 Gender

Female



33. Team Member 3 Contact Number

9080427125

34. Team Member 3 Engineering Branch

Information Technology



35. Team Member 3 Graduating Year

2028



## 4th Team Member Registration

36. Team Member 4 Name

Enter your answer

37. Team Member 4 Email Id

Enter your answer

38. Team member 4 Gender

Select your answer



39. Team Member 4 Contact Number

40. Team Member 4 Engineering Branch

Select your answer



41. Team Member 4 Graduating Year

Select your answer



## Project details

42. Project Title \*

Provide a concise title for your project that effectively captures its essence and purpose.

Edge AI for Vehicle Health and Predictive Maintenance

43. Please select project category under which you want to submit your project \*

Select the project category to which your project belongs from the provided list of choices. You can refer to the rules and regulations for further details on each category.

- ☒ AI at the Edge Solutions for Automotive
- ☐ AI at the Edge Solutions for Industrial Heavy Machinery
- ☐ AI at the Edge Solutions for Aerospace

44. Please select the project subcategory under the "AI at the Edge Solution for Automotive" category \*

Edge AI for Vehicle Health and Predictive ... 

45. Explain the specific real-world problem statement that your project is addressing \*

In this field, please provide a clear and concise explanation of the specific real-world problem that your project aims to address. Describe the issue or challenge your project seeks to solve or improve upon,

ensuring that it is well-defined and relevant to the chosen industry vertical and project category. Please share your response within 4000 characters.

Many vehicle breakdowns occur because faults are detected only after they become severe, leading to costly repairs, unexpected downtime, and safety risks. Traditional maintenance follows fixed service schedules rather than the actual condition of vehicle components. This project aims to continuously monitor vehicle health using Edge AI and onboard sensors to detect anomalies in real time and predict potential failures before they occur. This enables proactive maintenance, improves vehicle safety, reduces maintenance costs, and minimizes unexpected breakdowns.

46. Explain how your project addresses the problem statement and the value it creates. \*

Provide a detailed explanation of your proposed solution. Describe how it works, its key features, and the innovations involved, and explain how it addresses the problem statement. Highlight what makes your approach unique, practical, and effective. Please share your response within 4000 characters.

The proposed solution uses Edge AI deployed on an embedded device connected to vehicle sensors that collect real-time data such as engine temperature, battery voltage, vibration, fuel efficiency, and engine RPM. Machine learning models process this data locally to identify abnormal patterns and predict component failures without depending on cloud connectivity. Drivers receive instant alerts and maintenance recommendations, while service centers can access health reports for timely repairs. The solution provides faster decision-making, protects user privacy by processing data locally, reduces network latency, and ensures reliable operation even in areas with poor internet connectivity.

47. Explain the technologies/frameworks used in the above approach \*

Elaborate on the technologies and frameworks employed in your project's approach. Describe the specific tools, software, programming languages, or frameworks utilized to develop and implement your solution. Highlight any innovative or notable technological components that contribute to the effectiveness and functionality of your project. Please share your response within 4000 characters.

Python  
TensorFlow Lite / TensorFlow  
Scikit-learn  
Edge AI  
Raspberry Pi or NVIDIA Jetson Nano  
Arduino/ESP32 with vehicle sensors  
MQTT or CAN Bus for sensor communication  
Flask/FastAPI for dashboard API  
SQLite/Firebase for data storage  
HTML, CSS, JavaScript for dashboard visualization

48. Which AI category is your solution associated with? \*

☒ Edge AI

☐ Generative AI

- ☐ Edge AI / Agentic AI
- ☒ Predictive AI / Machine Learning
- ☐ Computer Vision / NLP / Speech AI
- ☒ Hybrid AI / Combination of multiple AI approaches

49. Describe the key benefits and impact that your project aims to deliver. How will it improve or address the current situation? (In terms of numbers, if possible.) \*

Highlight the key benefits and impact in quantifiable terms, which your project aims to bring to the current situation by addressing the identified problem. Please share your response within 4000 characters.

The project provides continuous vehicle health monitoring and predicts failures before they occur, helping vehicle owners and fleet operators avoid unexpected breakdowns. It can reduce maintenance costs by 20–30%, decrease unplanned vehicle downtime by 30–40%, improve maintenance planning, enhance passenger safety, and extend vehicle lifespan. Since AI processing is performed on the edge device, the system delivers low-latency predictions, reduces cloud dependency, and protects user data privacy.

50. What is the current state of your project? \*

Select the current state of your project from the provided dropdown options. Ensure that you support your selection with relevant supporting artifacts and a project progress plan in your project presentation.

Idea Deliberation Stage 

51. Does this project have Intellectual Property (IP) protection? \*

No 

52. If available, please mention the Patent Application Number, Patent Registration Number, Research Paper Title, DOI, Publication ID, or any other relevant reference number.

Enter your answer

53. Please confirm if this innovation project is entirely conceptualized by your team from the beginning and is not a direct copy or replication of an existing innovation or technology?

\*

Please confirm the conceptualization of the project by the team and share the findings from a benchmarking study of recent advancements in the project domain by other OEMs or startups. Additionally, explain what differentiates your project from the existing available solutions. Kindly provide your response within 4000 characters.

This project is entirely conceptualized and developed by our team. It is not a direct copy or replication of any existing product or technology. We studied existing vehicle monitoring and predictive maintenance solutions to understand their limitations, such as dependence on cloud connectivity and delayed fault detection. Based on this analysis, we designed an Edge AI-based solution that performs real-time vehicle health monitoring and predictive maintenance directly on the edge device. Our approach offers faster response, reduced latency, improved data privacy, and reliable operation even in low or no internet connectivity environments.

54. Have you filed or submitted this project in any other contests, competitions, or events previously? If yes, please mention the name of events where you have demonstrated this project. \*

Please mention if the project is presented in any other contest, competition, or events. Please share your response within 4000 characters.

No. This is the first time our team is submitting this project to a competition or innovation event.

## Participant Confirmation

55. Please confirm whether you have entered all the necessary information in the form and are content with the provided responses \*

Yes



56. How did you get to know about Tata Technologies InnoVent?

\*

College (Faculty, T&P cell)



57. Did you know about Tata Technologies as a company before participating in Tata Technologies InnoVent? \*

☒ Yes

☐ No

58. Add a Google Drive link to the folder with supporting documents: Detailed project presentation as per the template, supporting images and Videos of the POC and scanned copies of college ID card for all team members. \*

Please provide a Google Drive link to the folder containing the supporting documents for your project with full-view access, detailed steps are shared in the [Rules and Regulations \(Point 4.4 and 4.5\)](#) for the same. The folder should include the detailed project presentation in the specified format as shared in the Section 4 of [Rules and Regulations](#) along with supporting images and videos showcasing the current state of POC as specified.

Refer Rules and regulations: <https://www.tatatechnologies.com/tata-technologies-innovent-rules-and-regulations/>

[https://drive.google.com/drive/folders/1dDiqhruUwwC9Gnla4koQ\\_\\_iUU\\_GiriZ?usp=sharing](https://drive.google.com/drive/folders/1dDiqhruUwwC9Gnla4koQ__iUU_GiriZ?usp=sharing)



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